

POKHARA UNIVERSITY

Level: Bachelor
Programme: BCA
Course: Software Engineering (New)

Semester: Spring

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain Agile methods and highlight how Extreme Programming (XP) fits within Agile practices. 8
b) Explain design patterns. Describe the Singleton, Factory, and Observer patterns with examples. 7
2. a) Define encapsulation, inheritance, and polymorphism. Explain their significance in OOAD. 8
b) Why is code readability important in software development? Suggest techniques to improve readability. 7
3. a) Define Software engineering. List and Explain the Attributes of Good Software. 8
b) Define Requirement in Software Engineering. Differentiate between Functional and Non-functional Requirements. 7
4. a) Differentiate between verification and validation in software testing with suitable examples. 8
b) Why is software maintenance an essential phase in the software development life cycle? Analyze with suitable examples. 7
5. a) Draw a use case diagram for a Hostel Management System and describe the use case for 'Room Allocation'. 8

OR

Draw and explain a sequence diagram for a Train Ticket Booking System, Showing the flow from search to booking.

- b) What is Software Quality Assurance (SQA)? Describe the importance of SQA and explain any two quality standards. 7

OR

Explain Verification and Validation in software testing. How do they ensure software reliability?

6. a) Compare and contrast the Spiral model and Incremental Model. Which is more adaptable to changing requirements? 7
- b) What is software re-engineering? Explain how version control system (like GIT) help in managing software evolution. 8
7. Write short notes on: (**Any two**) 2×5
- a) RMMM Plan
- b) Human Computer Interaction
- c) Class diagram